

General Temperature Scale X

Class 11 Physics • Practice Questions

Name: _____

Date: _____

For every question, first identify **ICE**, **STEAM** and **CURRENT**.

LEVEL 1 — Direct conversion

1. A thermometer has ice point = 20°X and steam point = 80°X . What is the Celsius temperature when it reads 35°X ?
2. A scale X has $I_X = 20$ and $S_X = 80$. What will it read at 40°C ?

LEVEL 2 — Between two arbitrary scales

3. Scale X has $I_X = 10$, $S_X = 110$. Scale Y has $I_Y = 40$, $S_Y = 240$. Find the Y-reading corresponding to $X = 35^\circ\text{X}$.
4. For the same two scales as in Question 3, find the X-reading corresponding to $Y = 140^\circ\text{Y}$.

LEVEL 3 — Algebraic scales

5. On scale X the ice point is a and the steam point is $4a$. What is the Celsius temperature when the reading is $2a$?
6. On scale X the ice point is $2a$ and the steam point is $5a$. What is the Celsius temperature when the reading is $3a$?

LEVEL 4 — Negative values and readings outside the interval

7. Scale X has $I_X = -20$ and $S_X = 80$. Find the Celsius temperature when $X = 30$.
8. Scale X has $I_X = -10$ and $S_X = 90$. What Celsius temperature corresponds to $X = -60$?
9. Scale X has $I_X = 10$ and $S_X = 60$. What Celsius temperature corresponds to $X = 110$?

LEVEL 5 — Reverse problems

10. A scale X has a steam-point reading of 80°X . It reads 30°X when the actual temperature is 25°C . Find the ice-point reading.
11. A scale X has an ice-point reading of 10°X . It reads 40°X at 50°C . Find its steam-point reading.
12. Scale X has $I_X = -15$ and $S_X = 65$. Find the reading on X corresponding to 40°C .

THE KEY PHRASE

CURRENT – ICE, divided by STEAM – ICE